Experiment 1

Average Marks for Each Subject Using NumPy

Aim

To calculate the average marks for Math, Science, English, and History using NumPy arrays and identify the subject with the highest average score.

Procedure

1. Import the NumPy library.
2. Create a 4×4 NumPy array named student_scores containing the marks of four students in Math, Science, English, and History.
3. Use np.mean(student_scores, axis=0) to calculate the average marks for each subject.
4. Store the subject names in a list.
5. Use np.argmax() to identify the index of the highest average score.
6. Display the average marks of each subject and the subject with the highest average.

Code

```
import numpy as np

student_scores = np.array([
    [85, 90, 78, 88],
    [92, 85, 80, 75],
    [78, 88, 92, 80],
    [90, 95, 85, 90]
])

subjects = ["Math", "Science", "English", "History"]

avg_scores = np.mean(student_scores, axis=0)

print("Average scores:", avg_scores)

highest = np.argmax(avg_scores)

print("Highest average subject:", subjects[highest])

print("Highest average:", avg_scores[highest])
```

Result

Thus, the average marks for each subject were calculated using NumPy arrays, and the subject with the highest average score was identified successfully.

Output

```
======== RESTART: D:/fundamentails of data science/experiment 1 code.py ========
Average scores: [86.25 89.5  83.75 83.25]
Highest average subject: Science
Highest average: 89.5
```